An Isolated Fourier Shell Cannot Support a Nonzero L^2 Bound State

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Abstract

We prove that a square-integrable function on $\mathbb{R}^d$ whose Fourier transform is supported on a single wavenumber shell must vanish. The argument uses Plancherel's theorem and the zero d-dimensional Lebesgue measure of the shell. It gives a direct obstruction to nonzero L^2 eigenfunctions of the free-space Helmholtz operator and, more generally, of nonconstant polynomial functions of the Laplacian. A family of modulated wave packets then exhibits the separate freedom in envelope scale: fixing a carrier wavenumber alone does not select localization. The result concerns homogeneous, linear, whole-space spectral equations; it is not a no-go theorem for nonlinear bound states or bounded-domain modes.

1 Setting and main theorem

For $u \in L^2(\mathbb{R}^d;\mathbb{C})$ we use the unitary Fourier transform

$$\widehat{u}(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} u(x) e^{-ix\cdot\xi}\, dx, \tag{1}$$

first defined for $u \in L^1 \cap L^2$ and then extended by density and Plancherel to all of L^2. Write

$$S_{k_0} := \{\xi \in \mathbb{R}^d : |\xi| = k_0\}, \qquad k_0 \geq 0. \tag{2}$$

The support hypothesis below may be read either as a distributional support condition or as an essential-support condition on the L^2 representative; both yield the same conclusion.

Theorem 1 (Zero L^2 state on an isolated Fourier shell)**.** *Let $d \geq 1$, $k_0 \geq 0$, and $u \in L^2(\mathbb{R}^d;\mathbb{C})$. If*

$$\operatorname{supp} \widehat{u} \subseteq S_{k_0}, \tag{3}$$

then

$$\boxed{u = 0 \quad \text{almost everywhere on } \mathbb{R}^d.} \tag{4}$$

Proof. By Plancherel's theorem,

$$\widehat{u} \in L^2(\mathbb{R}^d), \qquad \|u\|_{L^2}^2 = \|\widehat{u}\|_{L^2}^2 = \int_{\mathbb{R}^d} |\widehat{u}(\xi)|^2\, d\xi. \tag{5}$$

We next show that S_{k_0} has zero d-dimensional Lebesgue measure. If $d = 1$, the shell contains at most the two points $\pm k_0$, so its Lebesgue measure is zero. If $d \geq 2$ and $k_0 > 0$, polar coordinates give

$$|S_{k_0}| = \int_{\mathbb{R}^d} \mathbf{1}_{\{|\xi|=k_0\}}\, d\xi \tag{6}$$

$$= \int_{\mathbb{S}^{d-1}} \int_0^\infty \mathbf{1}_{\{r=k_0\}} r^{d-1}\, dr\, d\omega = 0. \tag{7}$$

For $k_0 = 0$, the shell is the singleton $\{0\}$, which likewise has measure zero. Thus $|S_{k_0}| = 0$ in every dimension $d \geq 1$.

The support assumption in (3) implies that $\widehat{u} = 0$ almost everywhere off S_{k_0}. It follows that

$$\|\widehat{u}\|_{L^2}^2 = \int_{S_{k_0}} |\widehat{u}(\xi)|^2 \, d\xi = 0, \tag{8}$$

because $|\widehat{u}|^2 \in L^1(\mathbb{R}^d)$ and the domain of integration has measure zero. Equation (5) now gives $\|u\|_{L^2} = 0$, hence $u = 0$ almost everywhere. □

2 Consequences for homogeneous spectral equations

Corollary 2 (Free-space Helmholtz equation)**.** *For $d \geq 1$ and $k_0 \geq 0$, the distributional equation*

$$(-\Delta - k_0^2)u = 0 \quad \text{on } \mathbb{R}^d, \qquad u \in L^2(\mathbb{R}^d), \tag{9}$$

has only the trivial solution $u = 0$.

Proof. Fourier transformation in the sense of tempered distributions yields

$$(|\xi|^2 - k_0^2)\widehat{u}(\xi) = 0 \quad \text{in } \mathcal{S}'(\mathbb{R}^d). \tag{10}$$

On the open set $\mathbb{R}^d \setminus S_{k_0}$, the smooth multiplier $|\xi|^2 - k_0^2$ does not vanish. Local division by that multiplier in (10) shows that $\widehat{u}$ vanishes there as a distribution. Consequently $\operatorname{supp} \widehat{u} \subseteq S_{k_0}$. Theorem 1 then gives $u = 0$. □

Corollary 3 (Nonconstant polynomial dispersion)**.** *Let $P \in \mathbb{R}[t]$ be a nonconstant polynomial, let $\lambda \in \mathbb{R}$, and let $u \in L^2(\mathbb{R}^d)$ solve*

$$P(-\Delta)u = \lambda u \quad \text{in } \mathcal{S}'(\mathbb{R}^d). \tag{11}$$

Then $u = 0$ almost everywhere.

Proof. The Fourier-side equation is

$$(P(|\xi|^2) - \lambda)\widehat{u}(\xi) = 0. \tag{12}$$

Since $P(t) - \lambda$ is a nonzero polynomial, it has only finitely many real roots. Let $s_1, \ldots, s_N$ denote its nonnegative roots; the list may be empty. Local division outside the zero set of the multiplier shows

$$\operatorname{supp} \widehat{u} \subseteq \bigcup_{j=1}^{N} \{\xi : |\xi| = \sqrt{s_j}\}. \tag{13}$$

Each shell is Lebesgue null by the proof of Theorem 1. A finite union of null sets is null; if $N = 0$, the union is empty. Plancherel therefore implies $u = 0$. □

Remark 4 (General Fourier multipliers). The same reasoning applies to a Fourier multiplier $m(D)$ at a level λ whenever its eigenvalue equation forces the distributional support of $\widehat{u}$ into the set $\{\xi : m(\xi) = \lambda\}$ and that set has Lebesgue measure zero. The conclusion need not hold for multipliers that are constant on a set of positive Lebesgue measure.

3 Carrier wavelength versus envelope scale

The preceding theorem excludes a *nonzero exact-shell L^2 eigenstate*. It does not exclude localized wave packets with nonzero spectral bandwidth. The latter also show that selecting a carrier wavenumber does not independently determine an envelope radius.

Proposition 5 (Fixed carrier, freely varying envelope)**.** *Fix $k_0 > 0$ and a unit vector $\omega_0 \in \mathbb{S}^{d-1}$. Let $f \in \mathcal{S}(\mathbb{R}^d)$ be nonzero, with $\operatorname{supp} \widehat{f} \subset B(0, \rho)$ for some $\rho > 0$. For each $R > 0$ define*

$$u_R(x) = R^{-d/2} f(x/R) e^{ik_0 \omega_0 \cdot x}. \tag{14}$$

Then

$$\|u_R\|_2^2 = \|f\|_2^2, \tag{15}$$

$$\widehat{u}_R(\xi) = R^{d/2} \widehat{f}(R(\xi - k_0\omega_0)), \tag{16}$$

$$\operatorname{supp} \widehat{u}_R \subseteq B(k_0\omega_0, \rho/R), \tag{17}$$

$$\||x| u_R\|_2 = R \||x| f\|_2. \tag{18}$$

In particular, a fixed carrier k_0 and fixed mass permit a continuous family of envelope scales R.

Proof. For (15), change variables $y = x/R$ and use $|e^{ik_0\omega_0 \cdot x}| = 1$. Equation (16) follows from the Fourier scaling and modulation identities, and (17) follows at once from the support of $\widehat{f}$. Finally,

$$\begin{aligned} \||x| u_R\|_2^2 &= \int_{\mathbb{R}^d} |x|^2 R^{-d} |f(x/R)|^2 \, dx \\ &= R^2 \int_{\mathbb{R}^d} |y|^2 |f(y)|^2 \, dy. \end{aligned}$$

Taking square roots proves (18). □

From (17) and the reverse triangle inequality, all Fourier components in the packet support satisfy

$$||\xi| - k_0| < \frac{\rho}{R}. \tag{19}$$

Thus the spectral bandwidth narrows as $R \to \infty$, while the spatial extent grows linearly in R. A finite-width localized packet has spectral content beyond an exact shell. The examples u_R are kinematic constructions; they are not asserted to solve any particular evolution or stationary field equation.

4 Scope and interpretation

The mathematical conclusion is

$$\boxed{\operatorname{supp} \widehat{u} \subseteq \{\xi : |\xi| = k_0\}, \quad u \in L^2(\mathbb{R}^d) \quad \Longrightarrow \quad u \equiv 0.} \tag{20}$$

A nonzero distribution supported on a shell, such as a plane-wave Fourier delta or a surface measure, can define a generalized solution, but it is not a nonzero element of $L^2(\mathbb{R}^d)$.

The theorem does *not* rule out nonlinear localized states, externally trapped states, or modes on bounded and periodic domains. Those settings do not satisfy the exact whole-space, linear homogeneous shell-support premise. Nor does a chosen spectral minimum, by itself, establish a localized envelope, binding, stability, or confinement lifetime. Such conclusions require additional dynamical or variational hypotheses.

Related research audit: https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/cross_paper/LOCALIZATION_AND_DOMAIN_SELECTION.md.